

● EASY

● ALGEBRA

● ANSWER KEY

Linear functions

08

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Q1**D**

D. $y = 4.8x + 162$

Choice D is correct. It's given that the length of the whale was 162 cm when it was born and that its length increased an average of 4.8 cm per month for the first 12 months after it was born. Since x represents the number of months after the whale was born, the total increase in the whale's length, in cm, is 4.8 times x , or $4.8x$. The length of the whale y , in cm, can be found by adding the whale's length at birth, 162 cm, to the total increase in length, $4.8x$ cm. Therefore, the equation that best represents this situation is $y = 4.8x + 162$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q2**B**

B. 12

Choice B is correct. For the graph shown, the x -axis represents temperature, in kelvins, and the y -axis represents the estimated pressure, in pounds per square inch psi. The estimated pressure of the argon when the temperature is 600 kelvins can be found by locating the point on the graph where the value of x is equal to 600. The graph passes through the point 600, 12. This means that when the temperature is 600 kelvins, the estimated pressure is 12 psi.

Choice A is incorrect. This is the estimated pressure, in psi, of the argon when the temperature is 300 kelvins, not 600 kelvins.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the temperature, in kelvins, of the argon.

Q3**C****C.** 0, 2

Choice C is correct. The y-intercept of a graph is the point where the graph intersects the y-axis. The graph of $y = f(x)$ shown intersects the y-axis at the point $(0, 2)$. Therefore, the y-intercept of the graph of $y = f(x)$ is $(0, 2)$.

Choice A is incorrect. This is the x-intercept, not the y-intercept, of the graph of $y = f(x)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**C****C.** $f(x) = 9.25 + 0.50x$

Choice C is correct. It's given that the city's minimum hourly wage will increase by \$0.50 on the first day of the year for the 5 years after January 1, 2015. Therefore, the total increase, in dollars, in the minimum hourly wage x years after January 1, 2015, is represented by $0.50x$. Since the minimum hourly wage on January 1, 2015, was \$9.25, it follows that the minimum hourly wage, in dollars, x years after January 1, 2015, is represented by $9.25 + 0.50x$. Therefore, the function $f(x) = 9.25 + 0.50x$ best models this situation.

Choices A, B, and D are incorrect. In choice A, the function models a situation where the minimum hourly wage is \$9.25 on January 1, 2015, but decreases by \$0.50 on the first day of the year for the next 5 years. The functions in choices B and D both model a situation where the minimum hourly wage is increasing by \$9.25 on the first day of the year for the 5 years after January 1, 2015.

Q5**D****D.** 5

Choice D is correct. It's given that the function f is defined by $fx = \frac{1}{2}x + 6$. Substituting 4 for x in the given function yields $f4 = \frac{1}{2}4 + 6$, or $f4 = 5$. Therefore, the value of $f4$ is 5.

Choice A is incorrect. This is the value of $24 + 6$, not $\frac{1}{2}4 + 6$.

Choice B is incorrect. This is the value of $2 + 4 + 6$, not $\frac{1}{2}4 + 6$.

Choice C is incorrect. This is the value of $4 + 6$, not $\frac{1}{2}4 + 6$.

Q6**A****A.** -13

Choice A is correct. The value of $h-2$ can be found by substituting -2 for x in the equation defining h . Substituting -2 for x in $hx = 3x - 7$ yields $h-2 = 3-2 - 7$, or $h-2 = -13$. Therefore, the value of $h-2$ is -13.

Choice B is incorrect. This is the value of $h-1$, not $h-2$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**C**

C. $f(x) = 60x + 120$

Choice C is correct. It's given that the technician charges \$ 60 per hour for labor. Therefore, if the job takes x hours, the technician will charge $\frac{\$ 60}{1 \text{ hour}}x$ hours, or \$ $60x$, for labor. It's also given that the technician charges \$ 120 for parts. Therefore, $f(x) = 60x + 120$ represents the total amount, in dollars, the technician will charge for this job if it takes x hours.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This function represents the total amount, in dollars, the technician charges for labor only, not the total amount charged for labor and parts.

Choice D is incorrect. This function represents the total amount, in dollars, the technician would charge if the charge for parts were subtracted from, rather than added to, the charge for labor.

Q8**C**

C. $f(x) = 17$

Choice C is correct. An equation defining the linear function f can be written in the form $f(x) = mx + b$, where m is the slope and b is the y -intercept of the graph of $y = f(x)$ in the xy -plane. The slope of the graph of $y = f(x)$ can be found using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, where x_1, y_1 and x_2, y_2 are any two points that the graph passes through. If $f(0) = 17$, it follows that the graph of $y = f(x)$ passes through the point $(0, 17)$. If $f(1) = 17$, it follows that the graph of $y = f(x)$ passes through the point $(1, 17)$. Substituting $(0, 17)$ and $(1, 17)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{17 - 17}{1 - 0}$, which is equivalent to $m = \frac{0}{1}$, or $m = 0$. Since the graph of $y = f(x)$ passes through $(0, 17)$, it follows that $b = 17$. Substituting 0 for m and 17 for b in the equation $f(x) = mx + b$ yields $f(x) = 0x + 17$, or $f(x) = 17$. Thus, the equation that defines f is $f(x) = 17$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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